

Predicting Recommendation for Airports Based on Airport Reviews and determining key factors for Recommendation

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GitHub Repository: <https://github.com/lilyspy29-gif/CS3/tree/main>

Context

Today, the airport experience is at an all time low, with airline customers describing unpleasant trips, horrendous security lines, and airport ambiance at an all time low. With federal funding being spotty for airport operations, such as TSA, and the rise in jet fuel prices - airports need to find a way to maximize their margin in order to survive. As the majority of airport revenue is from operations and shopping, airports need to find new ways to improve their operations and their customer experience. With so many issues prevalent in the travel world today, how can we be so sure as to what customers care about the most? What will move the needle for customers to choose one airport over another?

In this project, we will use written airport reviews, rankings, and a binary recommendation factor to determine what makes an airport great. By using a logistic regression classifier and a Regression Model to determine feature importance, we can score and rank sentiment and help find key words that commonly come up with their associated ranking. The end goal will be to determine what words in a review are most likely to correlate with a recommendation or not a recommendation. By doing this, we can find key words to help determine areas of importance. As for the feature importance, it will help us figure out what aspects of the airport needs to be focused on first.

Task

You will be provided with a clean dataset containing customer reviews, customer origin, airport name, scores for different airport areas, and a binary recommendation column. Using this dataset, you are tasked with:

1. Create a Logistic Regression classifier model to classify the text - will then be used as inputs for TF-IDF
2. Create a Linear Regression model to determine feature importance of the airport areas

You will create visuals for the top 10 words that derive a "recommend", the top 10 words that derive a "not recommend", and a graph for showing feature importance for the areas of the airport.

Deliverable

Successful completion of this case study will include the following components:

1. Visuals for the written reviews and the airport areas - as listed above
2. Written conclusions from the data for each visual
3. A recommendation of how to help a given airport with their scores